PROGRAMMING IN JAVA

Assignment 01

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

Consider the following program.

```
class NPTEL {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

What is the output of the above code?

a. **Hello, World!**

b. **HelloWorld!**

c. **Compilation Error**

d. **Runtime Error**

Correct Answer: a

a. **Hello, World!**

Detailed Solution:

This is the standard syntax for a Java program to print the string "Hello, World!" to the console.

QUESTION 2:

A Java file with extension '.class' contains:

a. **Java source code**

b. **HTML tags**

c. **Java Byte code**

d. **A program file written in Java programming language**

Correct Answer: c

c. **Java Byte code**

Detailed Solution:

A .class file is a compiled version of the .java file in byte code. It is a kind of object code with the JVM (Java Virtual Machine) as the target machine.

QUESTION 3:

Which of the following is NOT an object-oriented programming paradigm?

a. **Encapsulation**

b. **Inheritance**

c. **Polymorphism**

d. **Dynamic memory allocation**

Correct Answer: d

d. **Dynamic memory allocation**

Detailed Solution:

Encapsulation, Inheritance, and Polymorphism are core paradigms. Dynamic memory allocation is a memory allocation strategy and not a programming paradigm.

QUESTION 4:

What is the correct sequence of steps to execute a Java program?

a. **Run → Edit → Compile**

b. **Edit → Run → Compile**

c. **Compile → Edit → Run**

d. **Edit → Compile → Run**

Correct Answer: d

d. **Edit → Compile → Run**

Detailed Solution:

The Java development process involves writing code (Edit), converting it to bytecode (Compile), and then executing it on the JVM (Run).

QUESTION 5:

Which program is used to compile Java source code into bytecode?

a. **javap**

b. **javac**

c. **java**

d. **javad**

Correct Answer: b

b. **javac**

Detailed Solution:

The JDK includes a tool called 'javac' that compiles from Java source code to a target of Java bytecodes. The program 'java' is used to execute Java bytecode.

QUESTION 6:

In Java, what is the role of the public static void main(String[] args) method?

a. **Initialization method**

b. **Execution entry point**

c. **Constructor**

d. **Destructor**

Correct Answer: b

b. **Execution entry point**

Detailed Solution:

The public static void main(String[] args) method is the entry point for the execution of a Java program.

QUESTION 7:

Which of the following is not a feature of Java?

a. **Platform Independence**

b. **Object-Oriented Programming**

c. **Supports Explicit Pointers**

d. **Supports Polymorphism**

Correct Answer: c

c. **Supports Explicit Pointers**

Detailed Solution:

Java is platform-independent, object-oriented, and secure. It does not support explicit pointers, as they can lead to memory management issues and compromise security.

QUESTION 8:

What is the primary purpose of the Java Virtual Machine (JVM) in the Java programming language?

a. **Graphical User Interface**

b. **Platform independence**

c. **Code Debugging**

d. **Game Engine**

Correct Answer: b

b. **Platform independence**

Detailed Solution:

The Java Virtual Machine (JVM) enables platform independence by interpreting Java bytecode, allowing Java programs to run on any device with a compatible JVM.

QUESTION 9:

Consider the following program.

```
public class Question {
    public static void main(String[] args) {
        int x = 5;
        x *= (2 + 8);
        System.out.println(x);
    }
}
```

What is the output of the above code?

a. 50

b. 10

c. 0

d. 5

Correct Answer: a

a. 50

Detailed Solution:

Here, x *= (2 + 8) is equivalent to x = x * (2 + 8), which results in x = x * 10. Therefore, x = 5 * 10 => x = 50.

QUESTION 10:

Which of the following is true regarding how Java handles code execution?

a. **Java uses only interpreter.**

b. **Java uses only compiler.**

c. **Java uses both interpreter and compiler.**

d. **None of the above.**

Correct Answer: c

c. **Java uses both interpreter and compiler.**

Detailed Solution:

Creating a .class file from .java using the javac command is a compilation task, whereas execution of a .class file using java is the process of interpretation.